

[illegible]

December 2025

(For VDT)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-



8. Each butterfly computation requires _____ complex multiplications and _____ complex additions.
- A) 3,2
B) 1,2
C) 2,1
D) 8,4
9. Long sequence filtering method that has aliasing effect is
- A) Overlap save
B) Impulse invariance
C) Overlap add
D) Linear filtering
10. A system where the passband cutoff frequency is greater than stopband cutoff frequency is
- A) Low pass filter
B) Band reject filter
C) Band pass filter
D) High pass filter
11. Filter design that plots all the poles of the filter on the unit circle is
- A) Chebyshev filter
B) Kalman filter
C) Butterworth filter
D) Elliptic filter
12. Identify the type of the filter that has the transfer function $|H(\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}}$
- A) Butterworth filter
B) Chebyshev filter
C) Elliptic filter
D) Kalman filter
13. Length of the impulse response of an IIR filter is _____
- A) Finite
B) Infinite
C) Cannot be determined
D) None
14. Task of obtaining a digital filter from the analog IIR filter involves developing a
- A) Mapping from s-plane to s-plane
B) Mapping from z-plane to z-plane
C) Mapping from z-plane to z-plane
D) mapping from s-plane to z-plane
15. The IIR filter transfer function contains
- A) Poles and zeros
B) Zeros only
C) Poles only
D) none
16. Bilinear transformation technique
- A) Transforms the entire left half of s-plane in to the unit circle in the z-plane
B) Transforms a strip on the left half of s-plane between $-\pi$ to π
C) Transforms a strip on the right half of s-plane between $-\pi$ to π
D) Transforms the entire right half of s-plane in to the unit circle in the z-plane
17. Which of the following windows has a time domain sequence $h(n) = 1 - 2 \frac{n - \frac{M-1}{2}}{M-1}$
- A) Rectangular window
B) Bartlett window
C) Blackman window
D) Hamming window
18. Which of the following is a method for implementing an FIR system?
- A) Direct form
B) Lattice form
C) Cascade form
D) All of these
19. The basic arithmetic operations will not include one of these
- A) ADD
B) AND
C) MULTIPLY
D) SUBTRACT
20. Which units are generally involved in Multiply and Accumulate (MAC)?
- A) Adder
B) Accumulator
C) Multiplier
D) All of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Using DFT-IDFT method, compute the circular convolution of $x_1(n) = (2, 1, 2, 1)$ and $x_2(n) = (1, 2, 3, 4)$.	6	L*3	1	1
b)	For a given sequence $x(n) = (0, 1, 2, 3, 4, 5, 6, 7)$ find DFT using DIT-FFT algorithm.	10	L3	2	1

2. a) A 14-point DFT of length 14 real sequence $x(n)$ is $X(k)$. The first 8 samples of DFT is given by

$X(0) = 12, X(1) = -1 + j3, X(2) = 3 + j4, X(3) = 1 - j5, X(4) = -2 + j2, X(5) = 6 + j3, X(6) = -2 - j3, X(7) = 10$. Find the remaining samples of $X(k)$. Also, evaluate the following.

(i) $\sum_{n=0}^{13} x(n)$ (ii) $\sum_{n=0}^{13} |x(n)|^2$

- b) Compute 8-point DFT of the sequence $x(n) = (1, 2, 3, 4, 4, 3, 2, 1)$ using DIF-FFT algorithm.

6 L3 1 1

10 L3 2 1

3. a) A length-8 sequence $x(n) = (-4, 5, 2, -3, 0, -2, 3, 4)$ has an 8 point DFT as $X(k)$. Determine the sequence $y(n)$ whose 8-point DFT is given by $Y(k) = W_8^{3k} X(k)$ using appropriate property.

6 L3 1 1

- b) Compute IDFT of the DFT samples $X(k) = (4, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.414, 0, 1 + j2.414)$ using DITFFT algorithm.

10 L3 2 1

Unit – II

4. a) Given two sequences $x(n) = (1, 2, -1, 2, 3, -2, -3, -1, 1, 1, 2, -1)$ and $h(n) = (1, 2)$ with a block length of 4. Perform long sequence filtering by using a method that has aliasing effect.

8 L3 3 1

- b) Realize the direct form-I and direct form-II structure for the system function

$$H(z) = \frac{z^{-1} - 3z^{-2}}{10 + 4z^{-1} + 4.5z^{-2} - 0.5z^{-3}}$$

8 L3 4 1

5. a) The normalized transfer function of a second order Butterworth filter is given by

$$H_2(s) = \frac{1}{s^2 + 1.414s + 1}$$

Convert the analog filter to a digital filter with cut-off frequency of 0.5π rad/sec using Bilinear transformation at $T=1$ sec.

8 L3 4 1

- b) Find the order, cut-off frequency and poles of a Butterworth LPF and plot the poles. The specifications are $\Omega_p = 20 \frac{\text{rad}}{\text{sec}}, \Omega_s = 30 \text{ rad/sec}, K_p = -2 \text{ dB}, K_s = -10 \text{ dB}$.

8 L3 3 1

6. a) Design a Butterworth low pass filter with PB attenuation as -1 dB and PB frequency as 4 rad/sec. SB attenuation is -20 dB with SB frequency as 8 rad/sec.

8 L3 3 1

- b) Convert an analog filter whose transfer function is

$$H_a(s) = \frac{4}{(s+2)(s+3)}$$

into a digital IIR filter using impulse invariance method with $T=2$ sec.

8 L3 4 1

Unit – III

7. a) Design a low pass filter using rectangular window sequence of length $M=11$ for the frequency response

$$H_d(e^{j\omega}) = 1; |\omega| \leq \frac{\pi}{2} \\ = 0; \frac{\pi}{2} \leq |\omega| \leq \pi$$

Obtain the impulse response and transfer function.

10 L3 5 1

- b) With a neat connection diagram explain Barrel shifter.

6 L3 5 1

8. a) Determine the impulse response of a FIR filter with reflection coefficients $K_1 = 0.6, K_2 = 0.3, K_3 = 0.5, K_4 = 0.9$ and $m = 4$. Also realize the direct form structure.

10 L3 5 1

- b) With a neat block diagram explain the operation of ALU.

6 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
